SET B Design and Analysis of Software Systems 30/01/2025

Quiz - 1

Name: ____________________________ **Roll No:** _____________

This is a closed-book, closed-notes Quiz. Answer all questions for a maximum score of **30 points**. The total time for the Quiz is **45 min**.
Credit is given for what you write, not what you are thinking. Write your answer in the space provided for the question. Partial credit will be given based on content, not quantity.

Good Luck!

Multiple choice questions (10 * 1 = 10 points)

1. **Which statement about the Spiral Model is incorrect?**
   A. It incorporates risk analysis as a fundamental part of its development process.
   B. It is particularly useful for large and complex projects with high-risk factors.
   C. It combines elements of both iterative and incremental development.
   D. It requires a fixed set of phases that must be completed in order.

   ANSWER: ______

2. **In which scenario would a Prototyping Model be preferred over a Waterfall Model?**
   A. When the requirements are well understood and unlikely to change.
   B. When the project involves multiple stakeholders with conflicting needs.
   C. When strict adherence to documentation and process is required.
   D. When there is uncertainty in user requirements and frequent feedback is essential.

   ANSWER: ______

3. **What is the primary limitation of using the Incremental Model in software development?**
   A. It requires customers to be involved continuously throughout the development process.
   B. It can lead to incomplete documentation if not properly managed.
   C. It does not support the use of automated testing tools effectively.
   D. It fails to accommodate changes after the initial requirements are established.

   ANSWER: ______

4. **What is the primary difference between the waterfall model and the evolutionary model?**
   A. The evolutionary model breaks product development into a series of releases.
   B. The evolutionary model allows you to change the sequencing and content of phases and activities as the project proceeds.
   C. The evolutionary model includes the concept of assessing risk and making go/no-go decisions.
   D. The evolutionary model allows you to go back to earlier phases to make improvements to previously generated documents and code deliverables.

   ANSWER: ______

5. **In the Agile SDLC model, what is the primary emphasis?**
   A. A strict and inflexible project schedule
   B. Extensive documentation and planning
   C. Continuous collaboration with customers and responding to change
   D. Completing all development work upfront before any testing begins

   ANSWER: ______

6. **In the Incremental Iterative Model, how are user requirements typically managed throughout the development process?**
   A. All user requirements are defined at the beginning and must remain unchanged until the project is complete.
   B. User requirements are continuously gathered, evaluated, and incorporated based on feedback received after each iteration.
   C. User requirements are rarely revisited, focusing instead on completing the project with minimal revisions.
   D. User requirements are documented in a single extensive document and not updated throughout the project.

   ANSWER: ______

7. **In a project using the Critical Path Method (CPM), an activity has the following attributes: Early Start (ES) = 4 days, Duration = 6 days, Late Finish (LF) = 12 days. What is the Late Start (LS) for this activity, and what does this indicate about the activity's float?**
   A. LS = 8 days; the activity has 4 days of float.
   B. LS = 6 days; the activity has 0 days of float.
   C. LS = 12 days; the activity has no flexibility in scheduling.
   D. LS = 10 days; the activity has 2 days of float.

   ANSWER: ______

8. Which of the following strategies is NOT primarily addressing the problems of concurrent development in a team environment?

   A. Strong focus on requirements elicitation and understanding of customer needs
   B. Effective configuration management procedures and tool use
   C. Good modular design with well-defined interfaces
   D. A clear integration strategy with good unit and integration testing

   ANSWER: ______

9. **In the context of software effort estimation, which of the following statements best describes the "cone of uncertainty"?**
   A. It suggests that as a project progresses, the estimation accuracy decreases significantly due to evolving requirements.
   B. It indicates that the initial estimation should always be considered the final benchmark for project budgeting.
   C. It emphasizes that the accuracy of project estimates improves as information becomes available over time.
   D. It illustrates that all estimation techniques yield approximately the same level of accuracy regardless of project complexity.

   ANSWER: ______

10. **When employing the Wideband Delphi technique for software estimation, what is the primary role of the facilitator?**
    A. Provide expert opinions on estimation values to the group.
    B. Ensure that all estimates are averaged into a single value for decision-making.
    C. Prepare the estimation report based solely on the individual estimates submitted by team members.
    D. Guide the team through the estimation process while managing group dynamics and ensuring all voices are heard.

    ANSWER: ______

11. **Explain the concept of "Cone of Uncertainty" in software estimation. How does it evolve throughout a project's lifecycle (3 points)**

12. **What are the triple constraints in project management? (Hint: Think about the triangle in drawn in class). Explain briefly how process influences the triple constraints. (4 points)**

13. **What are the key differences between Waterfall and Scrum software development methodologies, and when might you choose one over the other for a project? (3 points)**

14. **For the task table given below, perform the following (6 points)**:

Task Identifier	**Estimated Hours**	**Task Predecessors**
M	10	None (start)
N	25	M
O	20	M
P	20	O
Q	18	N
R	10	Q
S	52	P,R
T	30	Q
U	0 (done)	T,S

a. Construct a network diagram:

b. Identify the critical task path:

c. Assuming that all its preceding tasks are performed in exactly the hours estimated how much slack time does Task P have?

d. Assuming that all its preceding tasks are performed in exactly the hours estimated how much slack time does Task Q have?

15. (5 points)

A team is tasked with developing a customer relationship management (CRM) system for a medium-sized enterprise. The following specifics about the project have been established:

- The system requires core functionalities to be delivered in a series of increments, prioritized by business value.
- Key stakeholders are committed to providing feedback after each increment to refine and adjust future increments based on their experiences.
- The project has a timeline of one year, with expectations for updates and additional functionalities to be incorporated based on user feedback throughout the development period.
- There is a significant emphasis on maintaining quality and adhering to the requirements due to the critical nature of customer data handled by the CRM system.

Considering these specifics, identify the potential benefits and challenges associated with using the Incremental Iterative Model for this project:

A. **Benefits**: Allows for quick delivery of core functionalities, enables regular stakeholder feedback, and adapts to changing requirements.
Challenges: Ensuring proper integration of increments, maintaining consistency across the system, and potential difficulties with documentation due to rapid iterations.

B. **Benefits**: Provides a clear and structured process for delivering final features, minimizes risk, and ensures thorough testing at each phase.
Challenges: Delays in receiving feedback due to lengthy development cycles, limited flexibility in adjusting requirements, and difficulty accommodating user feedback post-release.

C. **Benefits**: Facilitates high levels of documentation to track changes and requirements thoroughly, ensures stable releases with each iteration.
Challenges: Lengthy project timelines due to extensive documentation requirements, limited interaction with stakeholders, and a rigid approach to changes.

D. **Benefits**: Allows for detailed upfront planning and a theoretically predictable project schedule, making resource allocation simpler.
Challenges: Risks associated with non-adaptability to changing environments and user needs, leading to potential misalignment with end-user expectations.

Select the answer that best summarizes the potential benefits and challenges of using the Incremental Iterative Model for the given project. **Justify your selection in detail !**